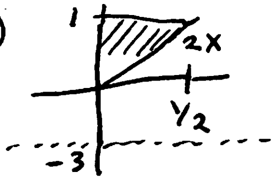
1452 Final Solutions – Fall 2014

(1)



(a) Shells $\int_0^1 2\pi (y+3)\left(\frac{1}{2}y\right)dy$

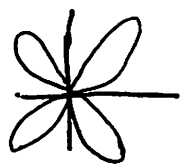
Washers $\int_0^{1/2} \pi(4)^2 - \pi(2x+3)^2\,dx$

(b) $\int_0^{1/2} (1-2x)^2\,dx$

(2)

r	θ
0	0
3	$\pi/4$
0	$\pi/2$

etc



$$\int_0^{\pi/2} \frac{1}{2}\left(3\sin(2\theta)\right)^2 d\theta$$

(3) Speed changes between $t=0$ and $t=4$ so the old static formula doesn't work. But we can divide the interval into pieces of length Δt. On each piece, the particle travels a distance $f(t_k)\Delta t$. Adding up these distances yields a Riemann sum. Taking the limit as $\Delta t \to 0$ produces the integral $\int_0^4 f(t)\,dt = \int_0^4 3t^2\,dt$.

(4) (a) $\int \frac{1}{x\sqrt{\ln x}}\,dx$ $\quad u=\ln x$, $du = \frac{1}{x}$ $\quad \int \frac{1}{\sqrt{u}}\,du = 2u^{1/2}+C = 2\sqrt{\ln x}+C$

(b) $\int e^{3x}\cos(2x)\,dx$ $\quad u = e^{3x}$, $du = 3e^{3x}dx$ $\quad dv = \cos(2x)\,dx$, $v = \frac{1}{2}\sin(2x)$

$\frac{1}{2}e^{3x}\sin(2x) - \frac{3}{2}\int e^{3x}\sin(2x)\,dx$ $\quad u = e^{3x}$, $du = 3e^{3x}$ $\quad dv = \sin(2x)$, $v = -\frac{1}{2}\cos(2x)$

$$\boxed{\int e^{3x}\cos(2x)\,dx} = \frac{1}{2}e^{3x}\sin(2x) + \frac{3}{4}e^{3x}\cos(2x) - \frac{9}{4}\boxed{\int e^{3x}\cos(2x)\,dx}$$

$$\frac{4}{13}\left(\frac{1}{2}e^{3x}\sin(2x) + \frac{3}{4}e^{3x}\cos(2x)\right) + C$$

4 (c) $\int \sin^3 x\,dx = \int \sin^2 x \;\; \sin x\,dx = \int (1-\cos^2 x)\sin x\,dx$

$u = \cos x$

$du = -\sin x\,dx$

$-\int 1-u^2\,du \qquad -\left(\cos x - \frac{1}{3}\cos^3 x\right) + C$

(d) $\int \frac{6x-4}{x(x+4)}\,dx = \int \frac{A}{x} + \frac{B}{x+4}\,dx \qquad A(x+4) + Bx = 6x-4$

$\underline{x=0} \quad 4A = -4 \quad A = -1 \qquad -\ln|x| + 7\ln|x+4| + C$

$\underline{x=-4} \quad -4B = -28 \quad B = 7$

5 $\int_2^3 \frac{5}{x-2}\,dx = \lim_{N\to 2^+} \int_N^3 \frac{5}{x-2}\,dx = \lim_{N\to 2^+} 5\ln|3-2| - 5|N-2| = \infty$

Diverges

6 (a) $\sum k^{1/k}$ diverges by Div. Test b/c $\lim_{k\to\infty} k^{1/k} = 1 \neq 0$

(b) $\sum \frac{\ln k}{\sqrt{k}} > \sum \frac{1}{\sqrt{k}}$ diverges by Direct Comp. Test

(c) $\sum \frac{3^k}{(2k)!}$ Conv. by Ratio Test $\lim_{k\to\infty} \frac{3^{k+1}}{(2k+2)!} \frac{(2k)!}{3^k} = \lim_{k\to\infty} \frac{3}{(2k+2)(2k+1)} = 0$

(d) $\sum \frac{1}{k^2+1}$ Conv. by Limit Comp Test $\lim_{k\to\infty} \frac{\frac{1}{k^2+1}}{\frac{1}{k^2}} = 1$

Also by Integral Test $\int_1^\infty \frac{1}{x^2+1}\,dx$ conv. to $\frac{\pi}{4}$

7 $\sum a_k$ Conv. abs. if $\sum |a_k|$ conv.

$\sum a_k$ Conv. Conditionally if $\sum a_k$ conv, but not absolutely

8 $\lim_{k\to\infty} \frac{a_{k+1}}{a_k} = \lim_{k\to\infty} \frac{3}{k} = 0 \implies \sum a_k$ Conv. by Ratio Test

(9) $f(x) = x\ln x$ $\qquad f(2) = 2\ln 2$

$f'(x) = \ln x + 1$ $\qquad f'(2) = \ln 2 + 1$

$f''(x) = \frac{1}{x}$ $\qquad f''(2) = \frac{1}{2}$

So $f(x) = 2\ln 2 + (\ln 2 + 1)(x-2) + \frac{\frac{1}{2}}{2!}(x-2)^2$

(10) $u = \langle 1, 4, -3 \rangle$ $\qquad v = \langle 2, 0, -1 \rangle$

(a) $u - 2v = \langle -3, 4, -1 \rangle$ $\qquad \|u - 2v\| = \sqrt{9 + 16 + 1} = \sqrt{26}$

So $\left\langle \frac{-3}{\sqrt{26}}, \frac{4}{\sqrt{26}}, \frac{-1}{\sqrt{26}} \right\rangle$ is unit vector

(b)

$u \cdot v = 1(2) + 4(0) + (-3)(-1) = 5 \neq 0$ So <u>not</u> orthogonal.